CLARENCE DE MEDEIROS

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EDUCATION

Worcester Polytechnic Institute - Worcester, MA **Expected May 2029**
Pursuing Bachelor of Science in Mechanical Engineering | GPA: 3.60 | Varsity Track and Field

Relevant coursework: Intro to CAD, Intro to ECE, Intro to Static Systems, Ordinary Differential Equations, Matrices and Linear Algebra, Intro to Materials Science, Applied Statistics

TECHNICAL PROJECTS & EXPERIENCE

Formula SAE - Drivetrain Subteam, Exhaust Lead **Aug 2026 - Present**

- Leading a 3-person exhaust team in designing and integrating the vehicle's exhaust system, including O2 sensor mounting and integration
- Developed preliminary exhaust routing CAD in SolidWorks to optimize packaging around engine and surrounding components
- Compiled bill of materials for O2 sensor, threaded bung, and custom pipe

High Power Rocketry Club - Experimental GNC Hardware Subteam **Aug 2026 - Present**

- Designing a gimbal test harness to simulate rocket motion across required degrees of freedom for ground testing
- Designed a full prototype gimbal in SolidWorks and presented it in PowerPoint at the club's Division Review
- Researched airbrake designs for the Mechanisms and Avionics subteam; proposed a peer university's design to combine coupling attachment and fin plate

WORK EXPERIENCE

Head Swim & Dive Coach - Woodland Golf Club (Newton, MA) **Jun 2025 - Present**

- Designed and implemented personalized daily training plans for 20+ divers
- Judged divers at competitions; created personalized diving sheets for each athlete
- Managed member/guest check-ins and pool and facility operations including inventory
- Served as a lifeguard and followed pool opening/closing procedures

Food Delivery Volunteer - Jewish Family & Children's Service **May 2023 - Present**

- Sort and deliver groceries, hygiene products, and essentials to 3-4 families biweekly
- Drove 1+ hour each trip to serve families in an outlying neighborhood

Crew Member - Dunkin' Donuts (West Newton, MA) **Jun 2023 - Aug 2025**

- Received, labeled, and organized inventory deliveries
- Operated the cash register and prepared food and beverages

SKILLS & CERTIFICATIONS

Technical Skills: Computer-Aided Design (CAD), Finite Element Analysis (FEA), Bill of Materials (BOM), 3D Printing, 3D Scanning
Software: SolidWorks, Onshape, MATLAB, RStudio, OrcaSlicer, Artec Studio Professional, Microsoft PowerPoint, Google Sheets, Excel
Certifications/Awards: ARC Introduction to MATLAB Certificate, Certificate of Engineering (Newton North High School), National African American Recognition Program Scholar, Price Meropol McMahon Award (Woodland Golf Club), Lifeguarding & CPR/AED for Professional Rescuers
Languages: English (fluent), French (native), Spanish (conversational)

ADDITIONAL INFORMATION

Activities: Track and Field (High jump, Triple Jump, Sprints), Lambda Chi Alpha Initiated Member